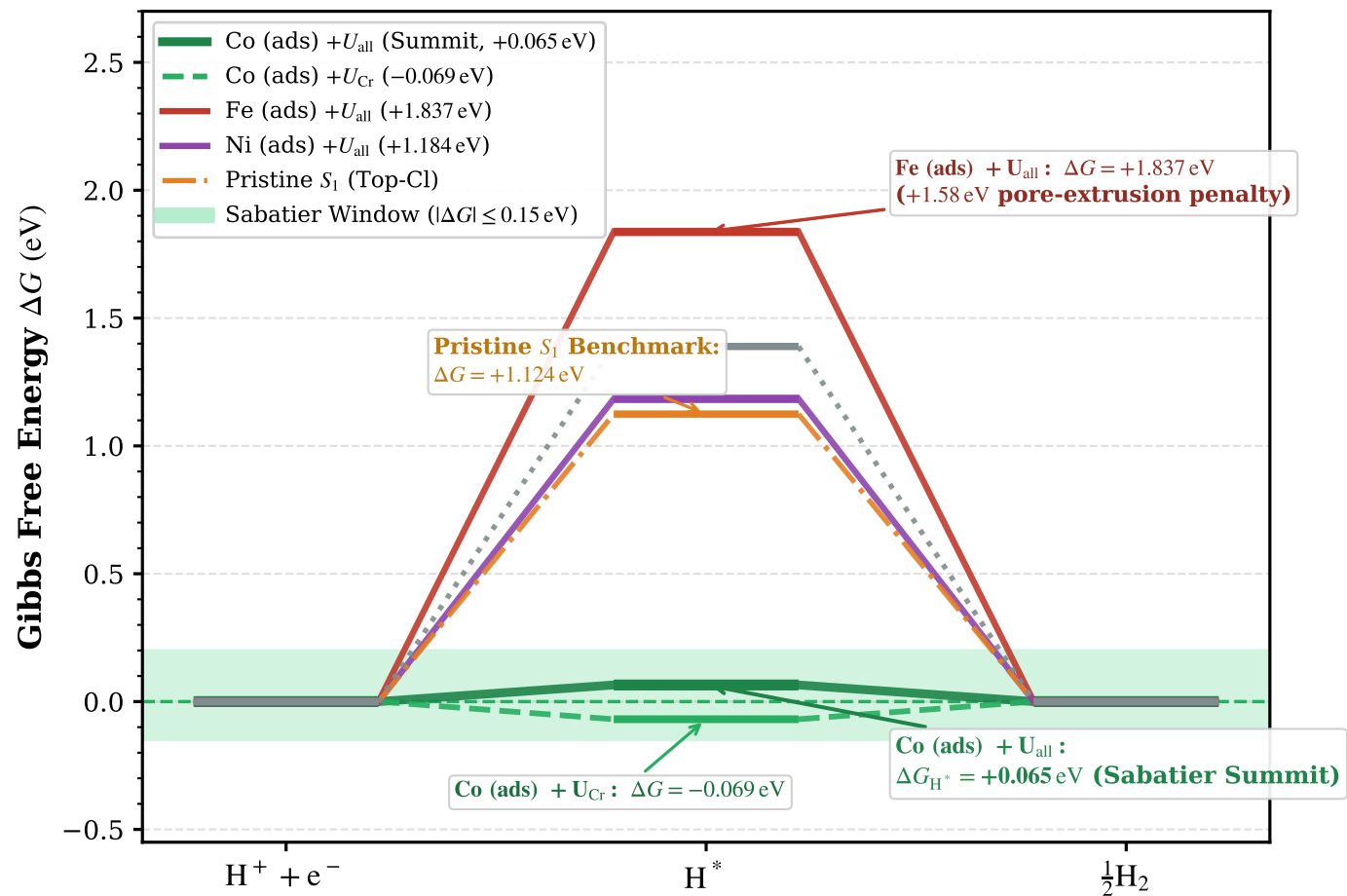
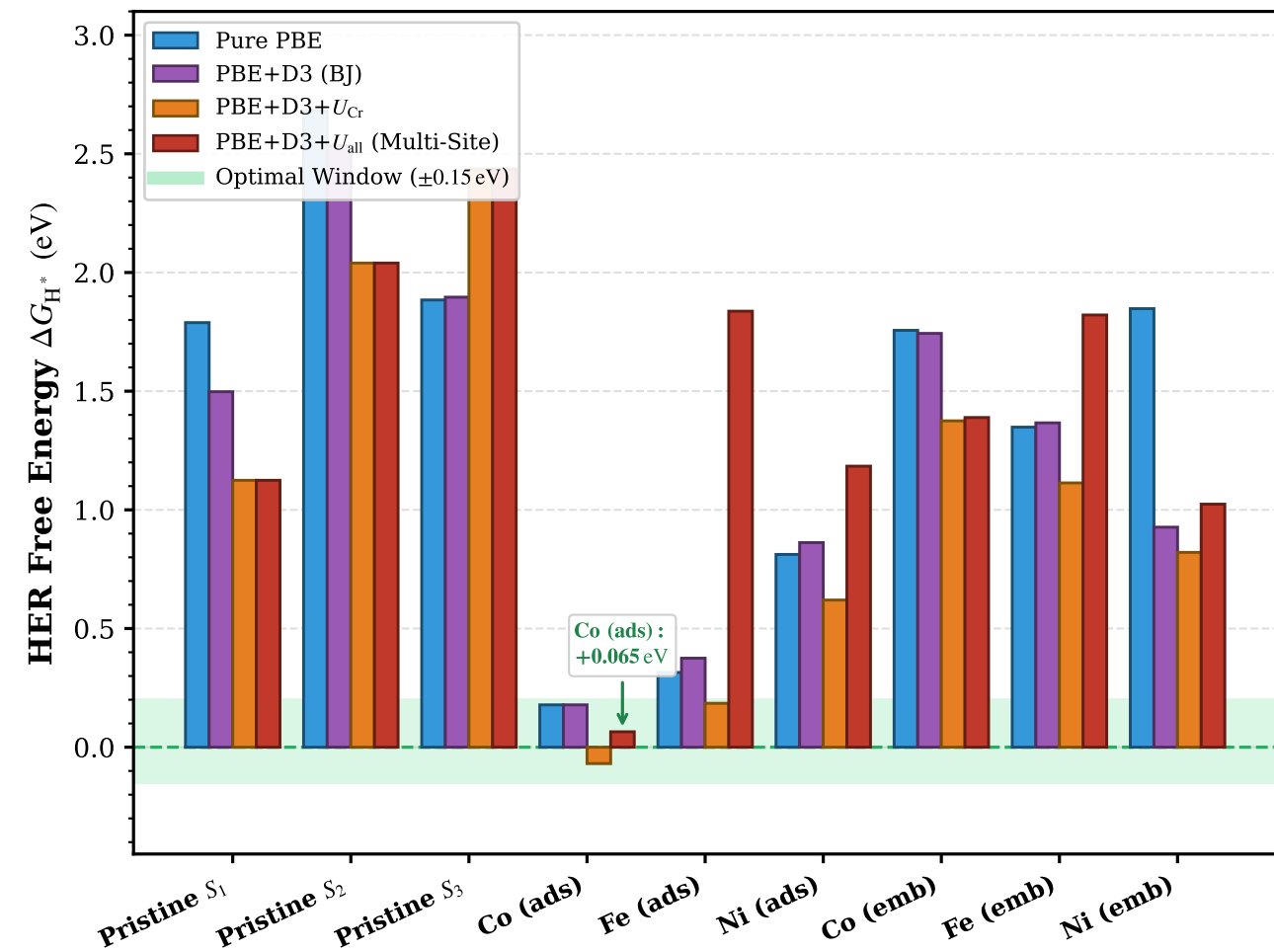
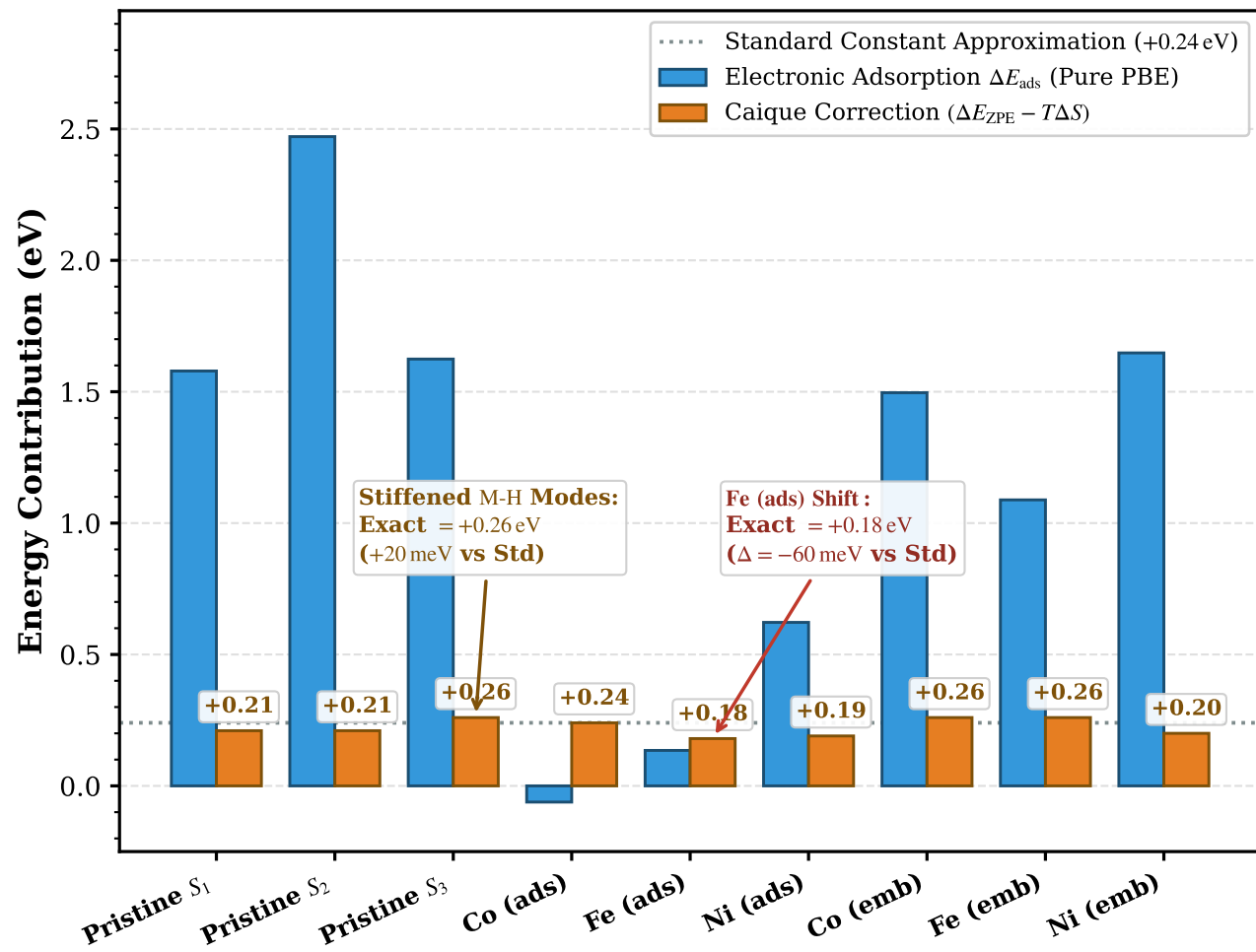


(a) HER Free Energy Reaction Diagram ($T = 298.15$ K)(b) 4-Tier Evolution of ΔG_{H^*} with Exact Caique Corrections(c) Energy Decomposition: Electronic ΔE_{ads} vs. Exact Caique Vibrational Offset(d) Methodological Energy Shifts ($\Delta\Delta G$) across All Tiers